Assignment - 2

① (1) (a) ~~$\hat{H} = \hat{p}^2$~~ ~~$\forall t$~~ $F = -\frac{dV}{dx}$

$\Rightarrow \int dV = -\int F\,dx$

$\Rightarrow V(x) = -Fx + c$

~~TISE~~ : ~~TDSE~~ : By convention, we take $c = 0$ as it only affects the energy eigenvalues

~~$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$~~ ~~$i\hbar\frac{\partial\psi}{\partial t}$~~

~~Let~~ ~~$\psi(x,t) = \phi(x)e$~~ Ex : H_1 $V_1 = -Fx$.

$V_2 = -Fx + c$.

$\hat{H}_1 = \frac{\hat{p}^2}{2m} + V_1 = \boxed{\frac{p^2}{2m} - Fx}$ → (Ans)

$\hat{H}_2 = \frac{p^2}{2m} + V_2 = \frac{p^2}{2m} - (Fx + c)$ $\quad \hat{H}_2 = \hat{H}_1 + c$

$\therefore H_1|\psi\rangle = E|\psi\rangle$

$\therefore H_2|\psi\rangle = (H_1 + c)|\psi\rangle = (E + c)|\psi\rangle$.

The eigenstates remain the same

$$\boxed{\frac{d\hat{A}}{dt} = \frac{1}{i\hbar}[A, \hat{H}] + \frac{\partial A}{\partial t}}$$ — Heisenberg's Eqn

$\therefore$ Putting $\hat{A} = x$: $\frac{d\hat{x}_H}{dt} = \frac{\hat{p}_H}{m}$ — ①

Putting $\hat{A} = \hat{p}$: $\frac{d\hat{p}_H}{dt} = -\frac{\partial V(\hat{x}_H)}{\partial x} = F$

from previous Tutorial-4 (Q1) calculat.

$\Rightarrow \boxed{\hat{p}_H = \hat{p}_0 + Ft}$ — (ii).

Replacing in ①:

$d\hat{x}_H = \frac{1}{m}(\hat{p}_0 + Ft)\,dt$

$\Rightarrow \boxed{\hat{x}_H = \hat{x}_0 + \frac{\hat{p}_0 t}{m} + \frac{Ft^2}{2m}}$ — (iii).

(b) $\langle x_H(t)\rangle = \langle \psi | x_H(t) | \psi \rangle$

$$= \langle \psi | x_0 + \frac{p_0 t}{m} + \frac{Ft^2}{2m} | \psi \rangle$$

$$= \underbrace{\langle \psi | x_0 | \psi \rangle}_{\langle x_0 \rangle} + \frac{1}{m}\langle \psi | p_0 t | \psi \rangle + \frac{F}{2m}\langle \psi | t^2 | \psi \rangle.$$

$$\langle \hat{x}_0 \rangle = \int_{-\infty}^{\infty} \psi^* x_0 \psi \, dx$$

$$= N^2 \int_{-\infty}^{\infty} e^{-\frac{x^2}{\Delta^2}} \cdot x \, dx = 0 \quad [\text{odd integral}].$$

Similarly, $\langle p_0 \rangle = 0$.

$$\langle \psi | p_0 t | \psi \rangle = t \langle \psi | p_0 | \psi \rangle \quad [\psi \text{ is time-independent}]$$

$$= t \langle p_0 \rangle = 0$$

$$\therefore \langle x_H(t) \rangle = \cancel{\langle \psi | \frac{Ft^2}{2m} | \psi \rangle = \langle \frac{Ft^2}{2m} \rangle} =$$

$$= \frac{F}{2m} t^2 \langle \psi | \psi \rangle = \frac{Ft^2}{2m} \quad [\psi \to \text{time-independent}]$$

$$\therefore \boxed{\langle x_H(t) \rangle = \frac{Ft^2}{2m}}$$

It travels in time like a classical particle with acceleration $a = \frac{F}{m}$

$$\left[x = ut + \frac{1}{2}at^2 + x_0 \; ; \; x_0 = 0 \; ; \; u = 0 \right]$$

(11) (a) $\rightarrow$ Solved using calculations of Tut-4 (Q1)

(b) $$\frac{d^2 x_H}{dt^2} + \frac{1}{m}\frac{\partial V}{\partial x}(x) = 0 \quad [\text{from Tut-4 (Q1) - (3)}]$$

$$\frac{\partial V}{\partial x} = kx = m\omega^2 x$$

$$\therefore \boxed{\frac{d^2 x_H}{dt^2} + \omega^2 x_H = 0} \quad \text{— SHM Eqn}$$

General Solution : $x_H(t) = A\cos\omega t + B\sin\omega t$

$$\frac{dx}{dt} = -\omega A \sin\omega t + \omega B \cos\omega t.$$

@ $t=0$: $x_H(0) = \boxed{A = \hat{x}(0)}$

$$\frac{dx_H}{dt}(0) = \omega B = \frac{\hat{p}(0)}{m} \quad (\text{from (3)}). \Rightarrow \boxed{\hat{p}(0) = m\omega B}$$

$$\therefore \boxed{\hat{x}_H(t) = \hat{x}(0)\cos\omega t + \frac{\hat{p}(0)}{m\omega}\sin\omega t} \quad \text{— (4)}$$

$$\therefore \frac{\hat{p}_H(t)}{m} = m\frac{d\hat{x}_H}{dt} = +m\omega(-A\sin\omega t + B\cos\omega t).$$

$$= m\omega\left(-\hat{x}(0)\sin\omega t + \frac{\hat{p}(0)}{m\omega}\cos\omega t\right)$$

$$\boxed{\hat{p}_H(t) = -m\omega\hat{x}(0)\sin\omega t + \hat{p}(0)\cos\omega t} \quad \text{— (5)}$$

(c) $x_H(t) = e^{i\hat{H}t/\hbar}\,\hat{x}\,e^{-i\hat{H}t/\hbar}$

~~$x_H(t) = x$~~ Taking $A = \frac{i\hat{H}t}{\hbar}$ & $B = \hat{x}$.

$$\therefore x_H(t) = x + \left[\frac{i\hat{H}t}{\hbar}, x\right] + \frac{1}{2!}\left[\frac{iHt}{\hbar}, \left[\frac{iHt}{\hbar}, x\right]\right] + \frac{1}{3!}\left[\frac{iHt}{\hbar}, \frac{iHt}{\hbar}, \left[\frac{iHt}{\hbar}, x\right]\right] + \dots$$

$$\therefore \left[\frac{i\hat{H}t}{\hbar}, x\right] = \frac{it}{\hbar}[H, x] = \frac{it}{\hbar}\left(-\frac{i\hbar}{m}\hat{p}\right) = \frac{\hat{p}}{m}t$$

(from previous calculations).

$$\therefore \left[\frac{iHt}{\hbar}, \left[\frac{iHt}{\hbar}, x\right]\right] = \frac{it}{\hbar}\left[H, \frac{\hat{p}t}{m}\right].$$

$$\boxed{[H, x] = -\frac{i\hbar}{m}\hat{p}}$$

$$\boxed{[H, p] = +i\hbar kx}$$

$$= \boxed{\frac{it^2}{m\hbar}[H, \hat{p}]} =$$

$$= \frac{it^2}{m\cancel{\hbar}}\left(+i\cancel{\hbar}\frac{\partial V}{\partial x}\right) \quad \text{(from previous calc.)}.$$

$$= -\frac{t^2}{m}\cdot kx$$

$$\therefore \left[\frac{iHt}{\hbar}, \left[\frac{iHt}{\hbar}, \left[\frac{iHt}{\hbar}, x\right]\right]\right]$$

$$= -\left[\frac{iHt}{\hbar}, \frac{t^2}{m}kx\right]. = -\frac{it^3}{m\hbar}k[H, x].$$

$$= -\frac{it^3k}{m\cancel{\hbar}}\cdot\left(-\frac{i\cancel{\hbar}}{m}\hat{p}\right).$$

$$= -\frac{t^3}{m^2}k\hat{p}$$

$$\therefore \hat{x}_H(t) = \hat{x}_0 + \frac{\hat{p}}{m}t + \frac{1}{2!}\left(-\frac{t^2}{m}kx\right) - \frac{1}{3!}\frac{t^3}{m^2}k\hat{p} + \dots$$

$$= x_0\left(1 - \frac{\omega^2t^2}{2!} + \dots\right) + \frac{\hat{p}_0}{m\omega}\left(\omega t - \frac{t^3}{3!}\omega^3 + \dots\right)$$

$$\boxed{\hat{x}_H(t) = \hat{x}_0\cos\omega t + \frac{\hat{p}_0}{m\omega}\sin\omega t}$$

$[k = m\omega^2]$

Taking $\hat{B} = \hat{P}$ & $\hat{A} = \frac{i\hat{H}t}{\hbar}$

$$\hat{P}_H(t) = \hat{P}_0 + \left[\frac{i\hat{H}t}{\hbar}, \hat{P}\right] + \frac{1}{2!}\left[\frac{i\hat{H}t}{\hbar}, \left[\frac{i\hat{H}t}{\hbar}, \hat{P}\right]\right] + \frac{1}{3!}\left[\frac{i\hat{H}t}{\hbar}, \left[\frac{i\hat{H}t}{\hbar}, \left[\frac{i\hat{H}t}{\hbar}, \hat{P}\right]\right]\right] + \dots$$

$$\cancel{= \hat{P}_0 + \frac{it}{\hbar} \cdot i\hbar kx + \frac{1}{2!}\frac{t^2}{\hbar^2}(i\hbar)^2 kx + \frac{1}{3!}\left(\frac{it}{\hbar}\right)^3 [H, x](i\hbar)^2 k}$$

$$= \hat{P}_0 + -tm\omega^2\hat{x} + \frac{1}{2!}t^2 m\omega^2 x$$

$$\left[\frac{i\hat{H}t}{\hbar}, \hat{P}\right] = \frac{it}{\hbar} i\hbar kx = -tmkx = -m\omega^2 tx$$

$$\therefore \left[\frac{i\hat{H}t}{\hbar}, \left[\frac{i\hat{H}t}{\hbar}, \hat{P}\right]\right] = \cancel{\frac{t^2}{\hbar^2}[H,} \frac{it}{\hbar}[H, -tkx]$$

$$= -\frac{it^2 k}{\hbar}[H, x]$$

$$= -\frac{it^2 k}{\hbar}\left(-\frac{i\hbar}{m}\hat{P}\right)$$

$$= -\frac{t^2 k}{m}\hat{P} = -t^2\omega^2\hat{P}$$

$$\therefore \left[\frac{i\hat{H}t}{\hbar}, \left[\frac{i\hat{H}t}{\hbar}, \left[\frac{i\hat{H}t}{\hbar}, P\right]\right]\right] = \frac{it}{\hbar}\cdot\left(-\frac{t^2k}{m}\right)[H, P]$$

$$= -\frac{ik}{\hbar m^2}t^3 \cdot i\hbar kx$$

$$= \frac{k^2}{m^2}t^3 x = m\omega^4 t^3 x$$

$$\therefore \hat{P}_H(t) = \hat{P}_0 - m\omega^2 tx - \frac{1}{2!}t^2\omega^2\hat{P} + \frac{1}{3!}\omega^4 t^3 x + \dots$$

$$= \hat{P}_0\left(1 - \frac{\omega^2 t^2}{2!} + \dots\right) - m\omega^2\hat{x}_0\left(\omega t - \frac{\omega^3 t^3}{3!} + \dots\right)$$

$$\Rightarrow \boxed{\hat{P}_H(t) = \hat{P}_0\cos(\omega t) - m\omega\hat{x}_0\sin\omega t} \quad \text{(Ans)}$$

② (a) For Hermiticity:

$$\hat{O} = \hat{O}^\dagger$$

$$\therefore \hat{O}^\dagger = (|\phi\rangle\langle\psi|)^\dagger = |\psi\rangle\langle\phi| \quad \text{— (i)}$$

$$\therefore \hat{O}^\dagger|x\rangle = |\psi\rangle\langle\phi|x\rangle = \underbrace{\langle\phi|x\rangle}_{\text{scalar}}|\psi\rangle$$

$$\therefore \hat{O}|x\rangle = |\phi\rangle\langle\psi|x\rangle = \langle\psi|x\rangle\langle\phi| \quad \text{— (ii)}$$

(I) implies that $\hat{O}^{\dagger}|x\rangle$ is a multiple of $|\psi\rangle$

(II) " " $\hat{O}|x\rangle$ " " " $|\phi\rangle$

(I) & (II) $\Rightarrow$ $\boxed{|\psi\rangle = \lambda|\phi\rangle}$ $\Rightarrow$ $\boxed{\langle\psi| = \lambda^{*}\langle\phi|}$

$\therefore \hat{O} = |\phi\rangle\langle\psi| = |\phi\rangle\lambda^{*}\langle\phi| = \lambda^{*}|\phi\rangle\langle\phi|$. — (III)

$\therefore \hat{O}^{\dagger} = |\psi\rangle\langle\phi| = \lambda|\phi\rangle\langle\phi|$. — (IV)

(III) & (IV) $\Rightarrow$ $\lambda = \lambda^{*}$ $\Rightarrow$ $\lambda \in \mathbb{R}$.

Conditions : $|\psi\rangle = \lambda|\phi\rangle$; $\lambda \in \mathbb{R}$.

(b) $\hat{O}^{2} = (\lambda|\phi\rangle\langle\phi|)(\lambda|\phi\rangle\langle\phi|)$.

$= \lambda^{2}|\phi\rangle\langle\phi|\phi\rangle\langle\phi| = \lambda^{2}\langle\phi|\phi\rangle|\phi\rangle\langle\phi|$.

$= \lambda\langle\phi|\phi\rangle\hat{O}$.

A projection operator must satisfy:

(1) Idempotence : $P^{2} = P$.

(2) Hermiticity : $P^{\dagger} = P$.

For (1) : $\lambda\langle\phi|\phi\rangle = 1$ $\Rightarrow$ $\boxed{\lambda = \dfrac{1}{\langle\phi|\phi\rangle}}$

~~For (1)~~ (II) is already satisfied

$\therefore \boxed{\hat{O} = \dfrac{|\phi\rangle\langle\phi|}{\langle\phi|\phi\rangle}}$

(c) $P_1 = |\phi\rangle\langle\phi|$. ; $P_2 = |\psi\rangle\langle\psi|$.

Why?

$P_1|\phi\rangle = |\phi\rangle\langle\phi|\phi\rangle = |\phi\rangle$ [if ϕ is normalised].

~~Thus, P_1 projects the~~ Let $|x\rangle = A_1|\phi\rangle + B_1|\psi\rangle$.

$\therefore P_1|x\rangle = A_1|\phi\rangle\langle\phi|\phi\rangle + B_1\,0\,|\phi\rangle\langle\phi|\psi\rangle$.

$= A_1|\phi\rangle$ [if ψ is orthogonal to ϕ].

P_1 "picks out" the component of $|\phi\rangle$ from a vector.

It "projects" the given vector onto $|\phi\rangle$.

$$\hat{O} = \frac{\cancel{|\phi\rangle\langle\phi|}}{\cancel{1}} \quad \lambda_1 \, |\phi\rangle\langle\phi|\psi\rangle\langle\psi|$$

$$\lambda_1 \langle\phi|\psi\rangle \, |\phi\rangle\langle\psi| = \lambda \langle\phi|\psi\rangle \hat{O}$$

$$\therefore \hat{O} = \lambda P_1 P_2 \; ; \quad \text{where} \quad \lambda = \frac{1}{\langle\phi|\psi\rangle}$$

(proved)